

Article

Cooling Damage Characterization and Chemical-Enhanced Oil Recovery in Low-Permeable and High-Waxy Oil Reservoirs

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Abstract: The well productivity of high-waxy reservoirs is highly influenced by temperature changes. A decrease in temperature can cause the precipitation of wax from the crude oil, leading to a decrease in the formation's drainage capacity and a drop in oil production. In this study, the wax precipitation of crude oil is characterized by rheological properties tests and differential scanning calorimetry (DSC) thermal analysis. The wax damage characteristics of cores and the relative permeability curves at different temperatures were investigated through coreflood experiments. Furthermore, nanoemulsion is selected as a chemical agent for injection fluid. The nuclear magnetic resonance (NMR) scanning technique is used to investigate the effects of oil recovery enhancement at different pores by increasing temperature and adding nanoemulsion. By comparing the changes in T₂ spectra and the distribution pattern of residual oil before and after liquid injection, the results have shown that both increasing temperature and adding nanoemulsion have a significant effect on oil recovery. The improvement of micropores is less pronounced compared to macropores. The produced oil mainly comes from the large pores. When the temperature is lower than the crude oil dewaxing point temperature, there is a serious dewaxing plugging phenomenon in the pores. Additionally, by observing the pattern of residual oil distribution at the end of the NMR online drive, it is hereby classified into wax deposition retention type, weak water washing retention type, and immobilized type, each with its own distinct characteristics. Wettability alteration and interfacial tension reduction can help to improve the drainage capacity of high-wax oil reservoirs, which is the main mechanism of nanoemulsion for enhanced oil recovery. These findings are highly valuable for enhancing the comprehension of the impact of highly waxed crude oils on drainage capacity and the ultimate oil recovery rate, particularly in relation to wax precipitation deposition.

Keywords: low-permeability sandstone; high wax content; variable water injection temperature; NMR



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1. Introduction

High-wax content oil, also known as crude oil with a wax content greater than 10%, is characterized by high viscosity, high wax content, and a high freezing point and can be found worldwide [1]. The Zhetybai oil field has been under development for more than 60 years. Mostly composed of sandstone, this oilfield has been subjected to water injection for an extended period. With an average wax content of about 20.7%, ranging from 18.6% to 24.2%, the formation temperature of this oilfield is between 80.8 °C and 100 °C, while the paraffin saturation temperature ranges from 51 °C to 62 °C. High-wax crude oil is particularly vulnerable to temperature changes. Consequently, during drilling and production operations, if the temperature falls below the wax saturation temperature, wax

precipitation damage occurs, thereby affecting the drainage capacity of the reservoir. This, in turn, leads to decreased recovery rates in production wells with superior original physical properties and a decline in production rates [2]. Wax crystal precipitation can generally be divided into three stages: nucleation, growth, and aggregation. Based on the complex generation mechanism of this process and the influence of wax crystal precipitation on the rheological and viscosity–temperature properties of crude oil, plenty of studies have been conducted in this area [3–6]. In practical production, when the composition of crude oil, crude oil temperature, ambient temperature, flow rate, production time, and pressure deteriorate, wax crystals will precipitate from the crude oil. These wax crystals, along with the resin, asphaltene components, and other macromolecular components, form a complex network structure. As a result, the viscosity of crude oil increases, and the flow resistance during the production and transportation of crude oil becomes more challenging [7–11]. At the same time, different modes and rates of change in the external environment will result in different rates of wax crystal precipitation and wax crystal morphology [12–14].

From the aspect of experimental methods, wax damage in pores can be investigated via scanning electron microscopy (SEM), two-dimensional glass-etching microfluidic models, Computerized Tomography (CT) scanning, and Nuclear Magnetic Resonance (NMR) scanning. Among them, SEM images can clearly give feedback on the wax shape and the degree of adsorption of different minerals to wax crystals at a certain moment but cannot reflect the wax damage over a continuous time period [15]. The microfluidic model can be designed according to the pore and throat structure of the actual core, but it is difficult to simulate the drainage process of crude oil and the actual wax precipitation mechanism under the high-temperature and high-pressure environment in the reservoir. CT scanning can obtain the in situ fluid distribution in the core and can even capture the dynamic process of drainage. However, the drainage process requires more than two kinds of fluid staining, and the experimental image data involves centralized software analysis and programming, which makes the process relatively complicated. NMR can quantitatively distinguish the fluid distribution in different pores of the rock sample, and based on T_2 spectra and scanning imaging detection, it can do real-time continuous tracking and comparison of pore crude oil. At the same time, rock porosity, pore size distribution, permeability, capillary pressure, and permeability can be characterized. This technology is developing rapidly in the petroleum field [16]. In recent years, low-field NMR technology has been widely used in wax precipitation deposition studies. Zhao et al. [17] characterized wax crystallization processes and evaluated the performance of chemical additives at temperatures lower than the wax precipitation temperature based on NMR scans. Morozov et al. [18] fabricated a cold fingering model for wax deposition measurements and integrated it into a nuclear magnetic resonance probe to realize a non-invasive imaging study of the in situ wax deposition process, which can help to understand the fundamental processes behind the wax deposition phenomenon. Saxena et al. [19] performed the estimation of the wax content of petroleum fractions using NMR-derived structural parameters of petroleum fractions based on NMR correlation studies of ^1H . Savulescu et al. [20] presented an improved relaxation time for measuring the mobility of waxes, asphaltenes, and macromolecules of highly condensable oils, quantifying the number of wax molecules captured in the crystal network to analyze wax–wax and wax–inhibitor interactions. In addition, they quantified wax crystallization in crude oil systems based on freely induced decay (FID) and Carr–Purcell–Meiboom–Gill (CPMG) sequences.

Thanks to the non-invasive mapping capabilities of NMR technology, combined with significant advancements in NMR equipment, data processing, modeling, inversion, and measurement techniques over the past few decades, NMR has gained popularity in near-surface geophysics and geophysical core inspection [21]. Numerous studies have focused on core analysis techniques, including Clay swelling and mud infiltration, Porosity and saturation mapping, and Capillary pressure curve measurement using Magnetic Resonance Imaging (MRI) technology [22–24]. Li et al. have developed a comprehensive and robust Bayesian optimization framework for extracting intrinsic physical properties by integrating

pore-scale forward modeling and experimental measurements of macroscopic system responses [25]. Tsuru et al. have demonstrated the three-dimensional visualization of flow characteristics using Magnetic Resonance Imaging in a lattice cooling channel [26]. Carlos et al. have estimated in situ fluid properties by combining the interpretation of Magnetic Resonance measurements [27]. In the field of oil well production, Elsayed et al. have provided a comprehensive review of recent advances in nuclear magnetic resonance (NMR) measurements for near-surface characterization using laboratory, borehole, and field technologies. They have focused on improving the production rate of surface NMR (SNMR), enhancing petrophysical NMR models for hydraulic conductivity and vadose zone parameters, and understanding the scale dependency of NMR properties [28].

The flow regime of high-wax crude oil has attracted considerable research attention. Ring et al. [29] based their work on the deposition of solid paraffin particles in the pore space, leading to recurring wellbore damage problems in the reservoir. A simulation study of paraffin precipitation and dissolution convection processes was carried out. When the low-temperature foreign fluid invades the high-temperature reservoir, it causes cold damage to the reservoir by blocking the pore space with paraffin wax, resulting in a drastic decrease in reservoir permeability and oil production rate. Nie and Yang [30] established a numerical simulation model by introducing the phase transition equation and combining the two-phase oil–water drainage field. They obtained several typical characteristics of the cooling damage in the wellbore, near-well zone. Xie et al. [31] studied the effect of temperature change on the Changchunling oil reservoir through experiments. They found that the drainage of highly waxy crude oil is particularly sensitive to temperature change, and the radius of the pore throat and relative permeability decreases sharply with decreasing formation temperature. Zhou et al. [32] presented the results of the experimental analysis of coreflood for waxy crude oil at different temperatures, which proved the rationality of hot water flooding. And Wang et al. [33] used CT scanning technology to study changes in pore drainage flow paths in rocks with positive depositional rhythms at different injection temperatures, retention of highly wax-containing oil fractions in the cores, displacement efficiency, and distribution of residual oil.

The above NMR techniques are mainly applied to the study of wax precipitation of wax-containing crude oils. However, the NMR investigation of wax precipitation injury of waxy crude oil in the core pore environment is less, and further research needs to be carried out. Therefore, this study utilizes online NMR technology to investigate the damage to the core drainage channel caused by wax crystal precipitation, the change in recovery rate, and the characteristics of residual oil distribution at different times in highly waxy crude oil by changing the injection temperature and the type of injection fluid. At the same time, we carried out tests of wax precipitation rheology and viscosity–temperature characteristics of high-wax content crude oil and analyzed the drainage characteristics and cooling damage of wax precipitation. The experimental results have practical significance and reference value for high-wax reservoirs.

2. Test Samples and Methods

2.1. Experimental Materials

The experimental materials for this study were provided by the field, where core samples consisted mainly of fine sandstone, medium sandstone, and gravelly medium sandstone, which were hydrophilic sandstone cores. The basic properties of the core samples are listed in Table 1.

The experimental oil field produces crude oil known for its high wax content, which was pretreated by heating and allowing it to cool naturally [5]. This pretreatment ensured that the physical and chemical properties of the oil samples were not influenced by factors such as thermal history and shear history, making them consistent with the actual data in the field and reusable. The crude oil viscosity (50 °C) was measured at 23.29 mPa·s, and the crude oil density (20 °C) was measured at 0.883 g/cm³. The saturate, aromatic, resin, and asphaltene (SARA) fractions of crude oil were determined to be 45.06%, 8.23%,

5.52%, and 0.91%, respectively, using the column separation method with reference to the SY/T5119-2016 standard [34].

Table 1. Basic properties of core samples.

No.	Test Item	Diameter/cm	Length/cm	Porosity/%	Permeability/mD
1	Oil–water relative permeability test	2.449	5.00	18.12	19.27
2	Cooling damage experiment	2.449	5.00	16.90	18.81
3	Coreflood experiments monitored via NMR	2.451	4.90	16.90	21.58
4		2.454	4.97	18.76	18.81
5		2.469	4.98	16.11	19.35
6		2.453	4.97	18.06	20.70
7		2.455	4.93	17.94	21.25

The injection fluids used in the experiment included deuterium oxide (D_2O) and formation water. These fluids had a mineralization of 148,465 mg/l, were of the $CaCl_2$ type, and had a density of 1.1 g/cm³. D_2O is used for the NMR scans to shield the hydrogen signal of the aqueous phase. Normally, NMR signals are produced when hydrogen atoms are exposed to a magnetic field. However, the use of D_2O as a solvent can eliminate the influence of hydrogen atoms on the NMR signals [33]. Nanoemulsion (NE) is selected as the chemical additive for the injection fluid, which mainly consists of mixed surfactants (nonionic and anionic surfactant), cosurfactants (low carbon number alcohol), and oil phase (terpenoids). The details of the production method can be referred to in our previous studies [35,36].

2.2. Experimental Methods

2.2.1. Compositional Analysis of High-Wax Crude Oils

(1) Gas chromatography (GC) analysis of waxy crude oil

The chemical characteristics and flow properties at 25 °C of wax-containing crude oils vary greatly due to differences in group composition and carbon number distribution [37,38]. The full hydrocarbon carbon number distribution of the target crude oil was tested using high-temperature gas chromatography analysis in accordance with standard GB/T 30431-2020 [39]. The analysis began with an initial temperature of 35 °C, and the temperature was gradually increased to 425 °C at a rate of 15 °C/min. The temperature was then maintained at this level for a duration of 10 min. The column's flow rate was set at 30 mL/min. By calibrating the standard with nC_5 – nC_{40} , the content of each alkane was obtained. In addition, the external standard method was employed to determine the carbon number distribution of all hydrocarbons in the experimental oil samples.

(2) Differential Scanning Calorimetry (DSC) thermal analysis

DSC thermal analysis is widely used in the quantitative study of wax properties and wax deposition in crude oil. In this study, the wax deposition process of oil samples was tested using differential scanning calorimetry according to the industry standard SY/T 0545-2012 [40]. The heat flow density change curve with temperature was plotted to determine the wax content. By integrating the curve, the cumulative exothermic amount of wax deposition of oil samples could be obtained. Furthermore, the heat of crystallization of wax was assumed to be 210 J/g. Based on this, the cumulative amount of wax deposition of the oil samples could be calculated by taking the ratio of the two values. Using this ratio, the cumulative amount of wax precipitation of the oil sample can be calculated as well [41–44].

2.2.2. Rheological Experiments on Highly Waxy Crude Oils

The viscosity measurements were carried out using a German MASS-III high-temperature rheometer to investigate the viscosity–temperature and viscosity–shear rate properties of the

oil samples. Before the experiment, the oil samples underwent heat treatment in a water bath to eliminate their shear history and dissolve any wax precipitates that had formed at 25 °C. The temperature of the water bath was maintained at approximately 45 °C for a duration of 2 h. Following this, the oil samples were shaken, and 6 mL of each sample was taken for the test. The test temperature range varied from 20 °C to 100 °C, with a temperature change gradient of 2.5 °C per minute. Measurements were conducted after 8 min of constant temperature under each temperature gradient. Four shear rates were selected for analysis, specifically including 200, 150, 100, and 50 s^{−1}.

2.2.3. Cold Damage Assessment of Dewaxing in Highly Waxy Crude Oils

We employed the CFS-10000 multifunctional coreflood system to conduct single-phase (crude oil) coreflood experiments, enabling us to evaluate the formation damage caused by the wax precipitation of high wax-containing crude oil. Using wax-containing crude oil as the displacement agent, we measured the change in single-phase permeability at different temperatures. Four sets of experiments were conducted at different temperatures: surface temperature (25 °C), surface wax precipitation temperature (45 °C), reservoir wax precipitation temperature (65 °C), and oil reservoir temperature (90 °C). The experimental procedure is as follows: (1) The core was vacuumed and heated to 90 °C, and the injecting oil was maintained at 90 °C; (2) The crude oil was injected at a flow rate of 0.1 mL/min, and the pressure difference across the core was recorded; (3) When the pressure difference remained constant, the core sample and crude oil were cooled down to the next temperature. And the same procedure as step 2 was conducted.

2.2.4. Oil–Water Relative Permeability Experiments under Different Temperatures

To investigate the drainage characteristics of waxy crude oil in a porous rock medium, We conducted oil–water relative permeability test on the oil samples according to GB/T28912-2012 standard [45]. The unsteady-state method was selected to measure the relative permeability of the low-permeability cores. The natural core was vacuumed for 9 h in the oil–water phase infiltration experiment. It was then saturated with formation water and placed in the core holder. The heater was adjusted to reach the set temperature, at which point the oil was injected at a certain flow rate. After the device was no longer out of the water, the oil saturation was calculated. Subsequently, the core was displaced until the water content of the produced liquid was over 98%. The water and oil production were recorded at a certain time interval. The pressure difference data were collected and then used to calculate relative permeability using the fraction flow method.

2.2.5. Coreflood Experiments Monitored via NMR

The NMR technique can be used to determine core porosity, pore size distribution, and fluid saturation non-destructively and accurately. It can also be used with specific coreflood devices to evaluate EOR techniques, including fracturing, thermal fluid injection, gas injection, and even acidizing. In the theory of nuclear magnetic resonance, the elements and their isotopes are placed in a stable magnetic field, generating Larmor inductions. When a magnetic field pulse is applied to change the field, the nuclei in a high-energy state resonate and quickly return to the original low-energy state after the radiofrequency (RF) pulse is stopped, causing relaxation. It is well established that the amplitude of the NMR signal is directly proportional to the size of the core pores. By inverting the T₂ envelope spectra and identifying the peak at the transverse relaxation time, the pore volume distribution and the ratios of different core sizes can be accurately determined. This provides a means to intuitively detect and analyze the cores with various sizes.

$$\frac{1}{T_2} = \frac{1}{T_{2b}} + \frac{1}{T_{2d}} + \frac{1}{T_{2s}} \quad (1)$$

$$\frac{1}{T_{2s}} = \rho \frac{s}{v} \quad (2)$$

$$\frac{s}{v} = \frac{F_s}{r} \quad (3)$$

$$T_2 = Ar \quad (4)$$

The transverse relaxation time of the fluid, T_2 , is formed via three different relaxation mechanisms: the pore relaxation time T_{2b} , generated via dipole interactions between hydrogen protons in the pore fluid; the pore surface relaxation time T_{2s} , generated through interactions of pore surface atoms and diffusion in the internal magnetic field gradient; and the diffusion relaxation time T_{2d} , all in ms. Under a uniform magnetic field, the relaxation rates of these mechanisms differ. For a saturated single-phase fluid, the rock volume relaxation time and diffusion relaxation time are generally negligible compared to the surface relaxation time (T_{2s}). Therefore, T can be approximated to be equal to T_{2s} . In petrophysics, the pore surface relaxation rate (ρ) is measured in $\mu\text{m}/\text{ms}$, the pore-specific surface ($\frac{s}{v}$) is measured in μm^{-1} , where s is the rock pore variable area in μm^2 , and v is the pore volume in μm^3 [46]. F_s represents the dimensionless pore shape factor for different pore models. The pore radius is denoted as r in μm . By letting $A = 1/\rho F_s$ the surface relaxation rate and pore shape factor of all pores in the reservoir rock can be treated as constant values, resulting in a constant coefficient A . This implies that there is a linear relationship between T_2 and pore throat radius, where larger pores correspond to larger T_2 values.

The formula for calculating the recovery rate corresponding to different times and pore space can be determined by comparing the peak area in the oil-saturated condition. The peak area represents the total value of the H-atom signal in the measured sample. Furthermore, the recovery rate corresponding to the pore space in different time periods can be calculated due to the unique flow characteristics of high-waxed crude oil and the influence of the wax itself on flow suppression. Moreover, the magnitude of T_2 spectral changes varies with the application of different driving agents at different temperatures.

$$\eta_0 = \frac{s_0 - s_i}{s_0} \times 100\% \quad (5)$$

and

$$\eta_1 = \frac{s_{0j} - s_{ij}}{s_0 - s_i} \times 100\% \quad (6)$$

The formula includes several variables. s_0 represents the area between the T_2 spectral line and the X-axis when the rock sample is saturated with oil. s_i represents the area between the T_2 spectral line and the X-axis during the time period after displacement i . s_{0j} and s_{ij} represent the area between the T_2 spectral line and the X-axis within a certain aperture range after displacement of time i .

The MacroMR12-150H-I high-temperature, high-pressure, low-field NMR analyzer manufactured by Suzhou, China Niumag Analytical Instruments Co. Ltd. was used in this experiment. A visual representation of the device can be seen in Figure 1. The magnetic field strength is 0.3 T. The relaxation time T_2 spectrum was measured using the CPMG sequence, with a maximum value of 10,000.0 ms and a minimum value of 0.01 ms. A smoothing factor of 1 and a sampling frequency of 200 kHz were used in the measurement.

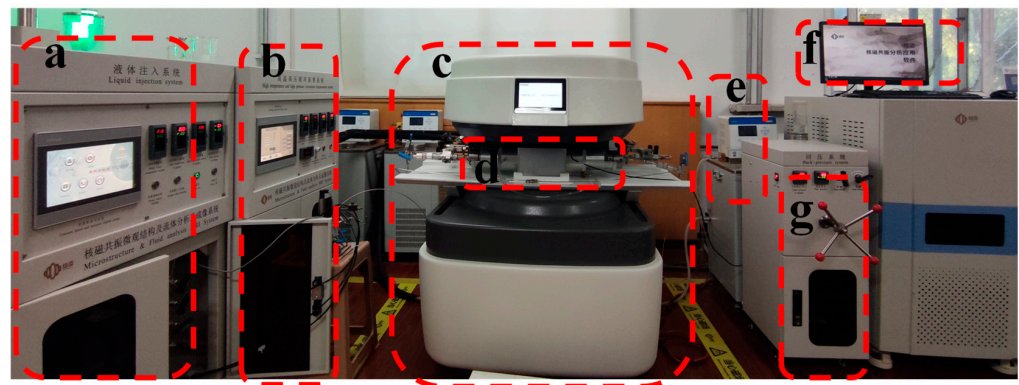


Figure 1. NMR-based online exfoliation system ((a) injection unit; (b) confining pressure unit (c) NMR unit (d) coreholder; (e) Heating unit; (f) Data acquisition unit; (g) Backpressure unit).

Five sets of experiments were designed, as shown in Table 2. These experiments were conducted at various temperatures (25 °C, 45 °C, and 65 °C) and different injection fluids (D₂O and nanoemulsion). The experimental procedure is as follows: (1) The core sample is placed into the coreholder and heated to the set temperature. (2) The injection fluid is also heated to the same temperature and then injected into the core at a certain flow rate. During this process, the T₂ spectra and magnetic resonance imaging (MRI) images are recorded at specific time intervals. (3) The experiment is concluded when the pressure difference reaches equilibrium. To simulate the reservoir environment, a confining pressure of 8 MPa was applied. The flow rate was set at 0.1 mL/min, and the nanoemulsion concentration was 0.05 wt%.

Table 2. NMR experimental schedule for core samples.

No.	Core Permeability (mD)	Temperature (°C)	Fluid Type
3	21.58	25	Formation water
4	18.81	45	Formation water
5	19.35	65	Formation water
6	20.70	25	0.05% nanoemulsion
7	21.25	65	0.05% nanoemulsion

3. Results

3.1. Wax Content Analysis of High-Wax Crude Oils

The measured crude oils' complete hydrocarbon carbon number distribution is illustrated in Figure 2. It is evident from the data in Table 3 that the measured crude oil exhibits a relatively uniform distribution and a low overall carbon number. The predominant components of the crude oil are those with a carbon number less than seven, indicating a minimal presence of heavy components. As crude oils are relatively viscous under ambient conditions, the composition of waxes plays a significant role in determining the crude oil viscosity.

Table 3. Carbon number classifications of crude oil.

Carbon Number	<C ₇	C ₈ –C ₁₅	C ₁₆ –C ₂₅	C ₂₆ –C ₃₀	>C ₃₀
Mass percentage (%)	18.89	38.36	24.79	11.53	6.44

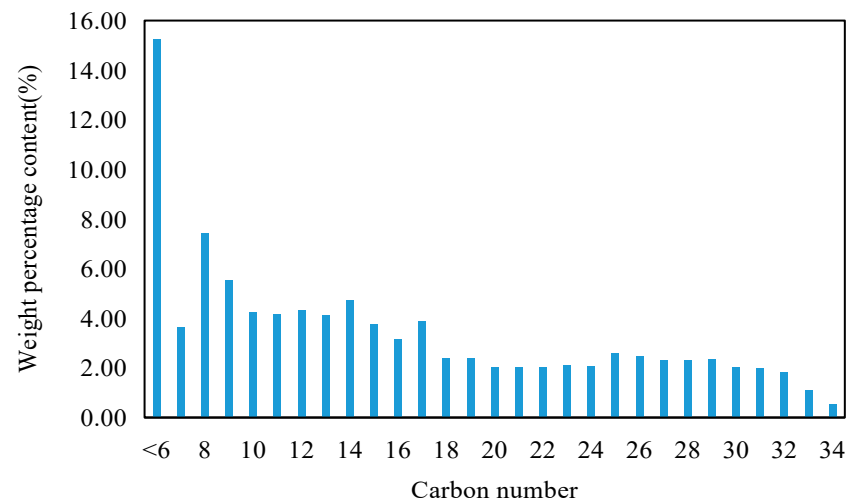


Figure 2. Carbon number distribution of crude oil.

Figure 3a indicates that the curve deviates from the baseline at 45.65 °C, suggesting the onset of wax precipitation. The cumulative wax precipitation curve of the crude oil, as depicted in Figure 3b, reveals a wax content of 24.93% in the oil sample.

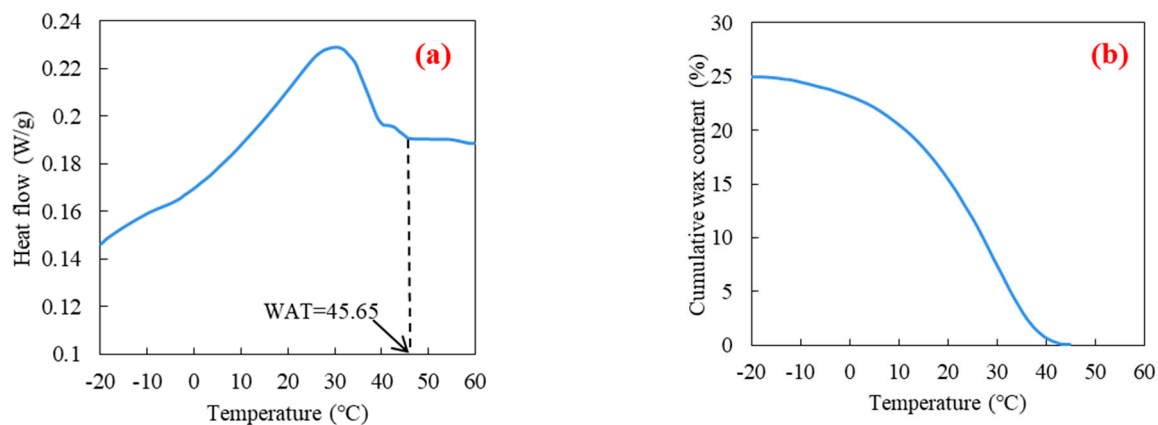


Figure 3. DSC test results of oil samples: (a) heat flow curve; (b) cumulative wax precipitation curve.

3.2. Rheological Properties of High-Wax Crude Oils

The viscosity of high wax-containing crude oil is sensitive to temperature change: under reservoir conditions (90 °C), the crude oil viscosity is 10.53 mPa·s, and under room temperature conditions (25 °C), the crude oil viscosity is as high as 900 mPa·s. The degree of change in the viscosity of the crude oil varies with different initial thermal histories, and it can be seen from the figure (Figure 4) that the change in the viscosity of the crude oil in the cooling process is higher than that in the warming process. When the crude oil temperature is higher than the wax precipitation anomaly point (35.6 °C), the temperature and shear rate have little effect on the crude oil viscosity, which shows Newtonian fluid characteristics. When the temperature is lower than the anomalous point of crude oil wax precipitation, the viscosity increases sharply with the decrease in temperature and shows non-Newtonian characteristics. The shear rate was in the range of 50 s⁻¹–150 s⁻¹, and with the increase in shear rate, part of the wax crystal network structure was sheared and destroyed via shear stress, improving the flow ability of the crude oil and leading to shear thinning. However, along with the further increase in the shear rate, unsteady flow occurs, and this unsteady flow forms elastic turbulence. When the shear rate reaches the elastic turbulence, the fluid not only does not continue to thin but also exhibits a shear thickening [47].

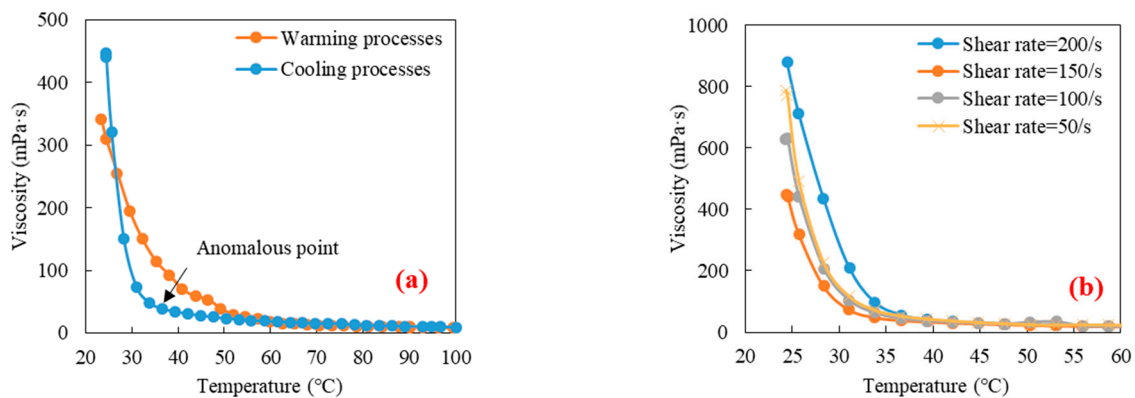


Figure 4. Crude oil viscosity–temperature curves: (a) viscosity–temperature curves for crude oil at 150 s^{-1} ; (b) viscosity–temperature curves at different shear rates.

3.3. Cold Damage Characteristics of Wax Analysis

The normalized permeability curve is shown in Figure 5. This demonstrates that as the drive progresses, the ratio of oil phase permeability ($K(t)$) to initial temperature permeability (K) steadily decreases. At temperatures above the wax precipitation point, the injection of crude oil and macromolecular substances can lead to the accumulation of blockages at the pore throat. This accumulation results in a decrease in rock permeability under these temperature conditions. At temperatures below the wax precipitation temperature, paraffin wax gradually precipitates out of the crude oil and forms aggregates, eventually leading to the plugging of the reservoir. It is observed that as the temperature decreases, the deposition of wax increases, consequently causing a greater drop in oil-phase permeability. For instance, when the temperature is $45\text{ }^{\circ}\text{C}$, the oil-phase permeability is reduced to 33% of the original oil-phase permeability observed at $90\text{ }^{\circ}\text{C}$. At $25\text{ }^{\circ}\text{C}$, the oil-phase permeability drops to only 11% of the original permeability observed at $90\text{ }^{\circ}\text{C}$. These results demonstrate that wax participation plays an important role in the hydrocarbon flow capacity of the porous medium.

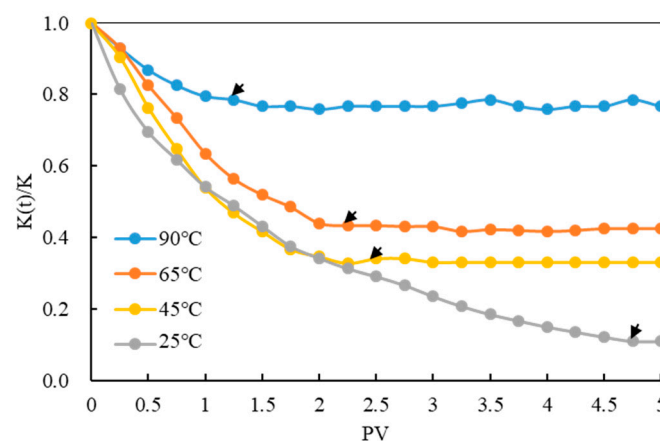


Figure 5. Normalized permeability curve changed with injection volume at different temperatures. (Oil-phase permeability basically ceases to change after the arrow).

3.4. Oil–Water Relative Permeability Characteristics

By assessing the relative permeability of oil and water at different temperatures and the degree of bound water saturation, we gained an understanding of the drainage characteristics of the oil and water phases in the porous medium. Figure 6 presents the measured oil–water relative permeability curves at different temperatures. It becomes evident that increasing the temperature leads to a gradual shift of the phase permeability curve to the right. Consequently, the two-phase co-drainage zone expands, and the isotonic

point water saturation increases. For instance, at 45 °C, the isotonic point saturation is 52%, while at 90 °C, it reaches 64%. As the temperature rises, the bound water saturation also increases, resulting in a significant decrease in residual oil saturation. Consequently, the two-phase co-drainage zone expands, thereby enhancing the drive efficiency. These changes mainly occur due to rise in temperature, which leads to a decrease in the viscosity of crude oil and a subsequent reduction in drainage resistance. Additionally, the wax crystals on the rock surface dissolve into the crude oil as the temperature increases. This phenomenon can significantly elevate the relative permeability of oil and water.

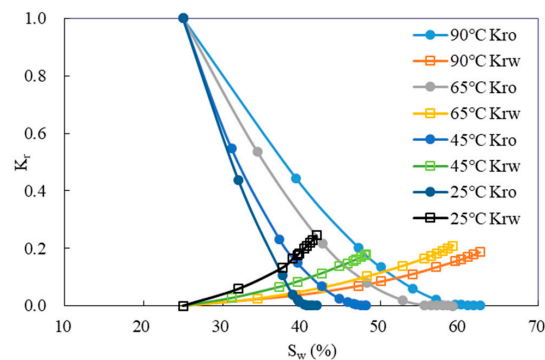


Figure 6. Oil–water relative permeability curves at different temperatures.

3.5. Coreflood Experiments Monitored via NMR

3.5.1. The Effects of Temperature

The T_2 spectra of low-permeability cores at different temperatures are shown in Figure 7. The T_2 spectrum of the 0 PV represents the oil-saturated state of the core sample. Overall, the saturated oil T_2 spectra exhibit a double-peak structure. By combining the T_2 spectra of the core samples with the findings of previous scholars [48], it can be inferred that regions with T_2 greater than 100 ms are deemed macropores, while regions with T_2 less than 10 ms are considered as micropores. At the surface temperature of 25 °C, the crude oil in the core primarily originated from the macropores, resulting in a significant decrease and shift in the T_2 spectra. However, the micropores showed weak changes due to their limited oil content. As the temperature increased to 45 °C, the crude oil saturation in the micropores increased due to differences in pore structures and the decrease in wax crystals and macromolecule content in the crude oil. This rise in temperature allowed the crude oil to permeate into even smaller pores. During the displacement process, the mobilization of crude oil was initially most obvious in the macropores and gradually slowed down, while the micropores were progressively mobilized until the end of the displacement. Furthermore, when the temperature was further increased to 65 °C, the T_2 spectra of macropores exhibited significant changes, characterized by a substantial decrease in spectral area and peaks. These changes were the main contributors to the mobilization of crude oil. Conversely, the spectral line changes in micropores were less pronounced.

When injecting the same pore volume (PV), the T_2 spectrum demonstrates different changes at different temperatures. At 25 °C, where the crude oil viscosity is high and difficult to flow, mobilization becomes challenging. In this case, injecting a 2 PV displacement fluid results in minimal changes to the spectrum of 0 PV, with only a slight decrease in peak intensity. Conversely, injecting hot water with 2 PV at 45 °C and 65 °C leads to a significant decrease in both the spectral area and peak intensity. The pressure differences in the cores during the displacement process are shown in Figure 8. Initially, under 25 °C, wax crystals precipitate and block the pores in the cores, leading to difficulty in displacement. This results in a pressure-holding process at 25 °C, where the differential pressure initially increases and then decreases. The maximum pressure difference is 2.64 MPa before reaching equilibrium. Conversely, when the temperature is above the wax precipitation point, the pressure difference reaches a peak and remains constant. Moreover, higher temperatures lead to smaller pressure differences between the two ends of the core, facilitating the driving

process. It can be noted that the equilibrium pressure difference under ambient conditions is twice as high as that under heating at 65 °C.

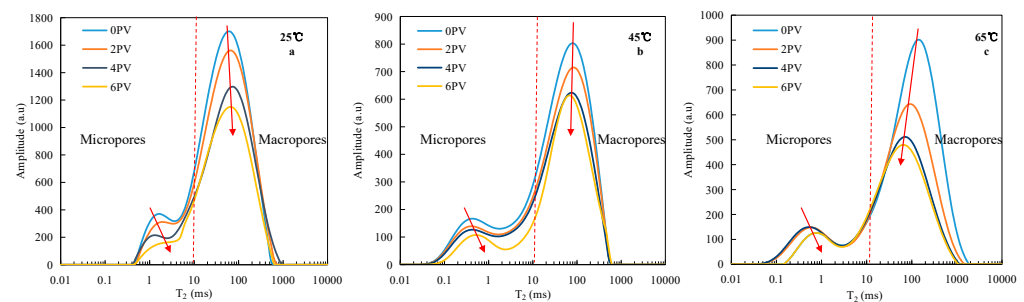


Figure 7. T₂ spectrum changed with PV at different temperatures: (a) 25 °C; (b) 45 °C; (c) 65 °C. (The red dashed line is the boundary between large and small pores, and the arrow is the direction of the peak change of the T₂ spectral line).

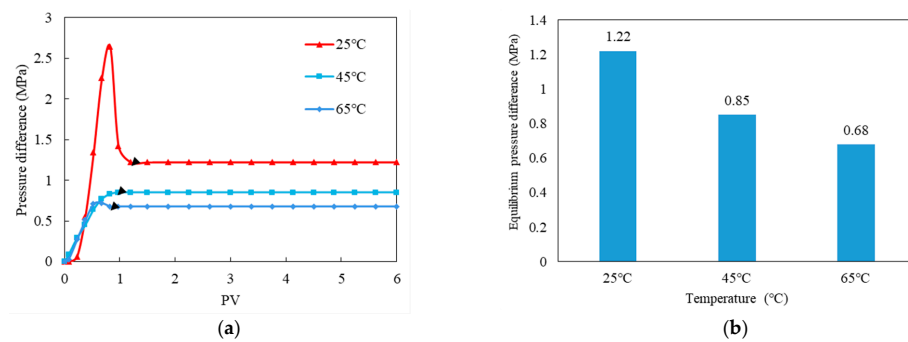


Figure 8. The pressure difference curves at different temperatures: (a) pressure difference curve; (b) equilibrium pressure difference. (Pressure equilibrium point is indicated by the arrow).

3.5.2. The Effects of Nanoemulsion

The T₂ spectra of adding 0.05% nanoemulsion at different temperatures are presented in Figure 9. It can be observed that the change in the spectral line under the same low permeability condition primarily occurs in macropores, indicating that the oil recovery is mainly contributed by the macropores. Adding nanoemulsion has a notable effect on the number of pore volumes required to achieve equilibrium compared to the water injection at the same temperatures. Specifically, under the temperature conditions of 25 °C, the number of pore volumes decreased by 1 PV. This demonstrates that the addition of nanoemulsion primarily enhances the oil recovery of the macropores.

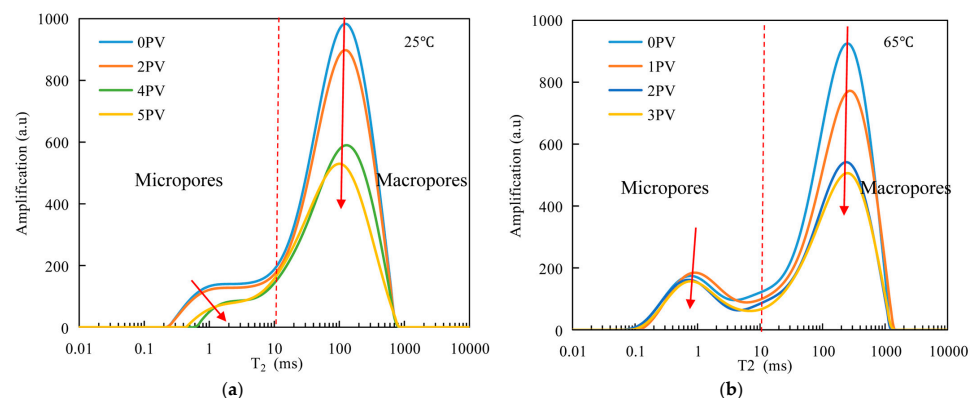


Figure 9. T₂ spectrum in nanoemulsion injection changed with PV at different temperatures: (a) 25 °C; (b) 65 °C. (The red dashed line is the boundary between large and small pores, and the arrow is the direction of the peak change of the T₂ spectral line).

3.5.3. The Comparison of Oil Recovery

Figure 10 summarizes the pore sizes and overall recovery rate of the core under different injection conditions. The overall recovery rate is similar to that of the macropores since the pore size distribution of the core is dominated by macropores. As the temperature increases, the wax crystal plugging is relieved, allowing the displacement agent to reach more pores and increasing the oil recovery rate. The recovery enhancement effect brought about by temperature increase is especially obvious for large pores. The oil recovery of macropores increases from 15.34% to 24.80% when the temperature is increased from 25 °C to 65 °C. Therefore, thermal improvement of oil recovery is particularly critical for high-wax crude oil reservoirs. Under low temperatures, the addition of 0.05% nanoemulsion to the injection fluid demonstrates a noticeable effect, increasing the overall recovery from 19.37% to 33.05%. However, when the temperature rises to 65 °C, the recovery enhancement effect by the nanoemulsion is less significant. At low temperatures, nanoemulsion can enhance the waxing behavior of wax molecules, inhibiting the growth and precipitation of wax crystals. Additionally, nanoemulsions easily infiltrates the micropore throat, improving the flow characteristics of crude oil in the core's roaring channel and effectively enhancing the oil-driving efficiency. As the temperature increases, the wax crystals dissolve, leading to improved crude oil flow and a less pronounced effect of nanoemulsion.

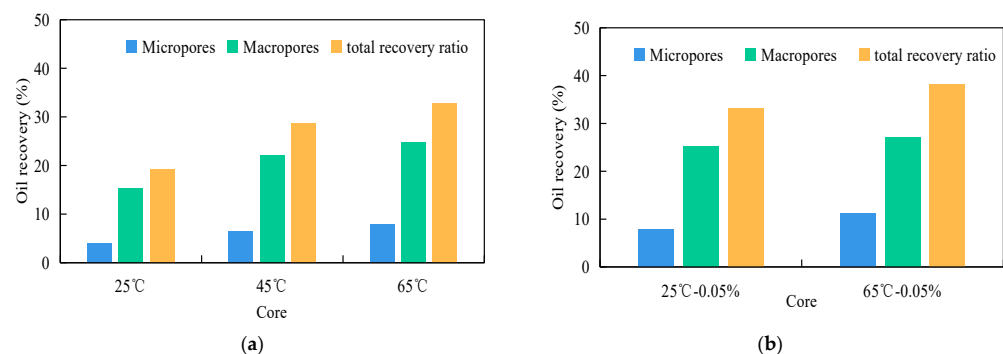


Figure 10. Comparison of oil recovery for different core samples: (a) water injection; (b) nanoemulsion injection.

As shown in Figure 11, the distribution pattern of the remaining oil after the displacement can be observed, which can be divided into wax deposition residual oil, weakly washed residual oil and unused residual oil. Wax deposition residual oil is mainly caused by low temperatures and mainly exists in most of the low-temperature ambient cores and at the inlet end of the high-temperature core displacement agent. Below the wax deposition point temperature, wax crystals precipitate out in the crude oil, forming a mesh structure that hinders the flow of crude oil and stagnates [49]. Figure 11a,b depicts the results of low-temperature displacements, showing darker and more contiguous remaining oil, mostly in the form of difficult-to-replace wax deposits. Furthermore, due to the injection of a normal temperature displacement agent, the crude oil cools down and deposits, resulting in a darker color at the inlet end of the rock sample compared to the outlet end. Weakly washed residual oil is mainly caused by insufficient washing and changes in the physical properties of the displacement agent. When the crude oil saturates into the core, it mainly enters the core pores and contacts the surface of the pore throat. The measurement results show that the core is in an oily-wet state. While the crude oil free in the inner part of the pore throat can be successfully replaced and eliminated by the foreign displacement agent, the crude oil adhered to the wall of the pore throat is difficult to remove, indicating inadequate displacement. From Figure 11c, it can be observed that the higher the displacement temperature, the less wax deposition residual oil and weakly washed residual oil. The addition of nanoemulsion can reduce the amount of weakly washed residual oil, as seen in the comparison between Figure 11c,e. The unused residual oil is mainly affected by the pore distribution of rock samples. It is difficult for the displacement agent to reach certain

areas of the rock samples, and dominant channels can interfere with the displacement. When the low-temperature displacement medium flows through the oil-bearing area, a wax film forms on the contact surface, increasing the flow resistance. In addition, when the crude oil content is high, or the oil quality is thick, a drive-in phenomenon occurs, where crude oil waxes and recombinant substances are adsorbed on the surface or inside the pore throat, hindering crude oil displacement. These conditions make it difficult for the injection fluid to reach certain areas and produce unused residual oil. Figure 11 shows scattered orange areas in individual cores as a result. Moreover, after prolonged driving, dominant channels form inside the core. Once the main flow channel is formed, the injection fluid (with low viscosity) has a large inertia force at high flow rates and moves toward the exit, forming scattered, isolated spots of unused oil. The lone points are more pronounced at higher injection temperatures [33].

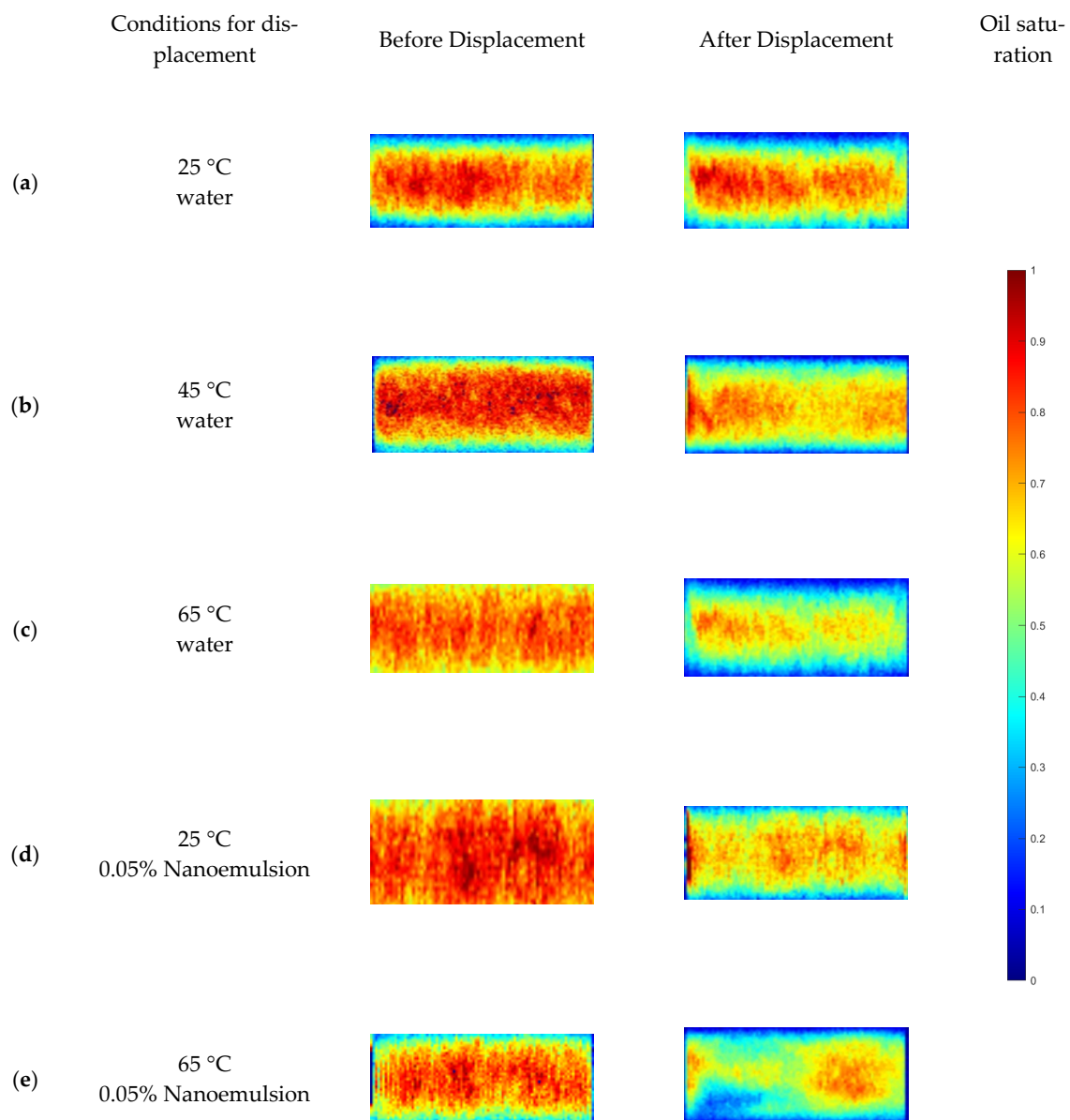


Figure 11. The MRI images for different core samples under different conditions. ((a) Distribution of residual oil in 25 °C water drive; (b) Distribution of residual oil in 45 °C water drive; (c) Distribution of residual oil in 65 °C water drive; (d) Distribution of residual oil in 25 °C nanoemulsion drive; (e) Distribution of residual oil in 65 °C nanoemulsion drive).

4. Discussions

The main reason for the difficulty of crude oil flow at low temperatures is the attachment of part of the remaining oil to the surface of the rock pores after the drive-off. Wax crystals precipitate partly attached to the surface of oil droplets, accompanied by some clay particles and fine mineral particles, which are wrapped in the crude oil. This is the main reason for the difficulty of crude oil flow at low temperatures. Figure 12b shows images obtained from a scanning electron microscope and demonstrates the different states of the remaining crude oil. In Figure 12a, the remaining crude oil is observed as droplets embedded within the rock pore interior. In Figure 12b, the remaining crude oil is completely spread and attached to the surface of the rock. These states are consistent with the weakly water-washed remaining oil described in Figure 11c–e. The former state can be effectively extracted by implementing a hot water drive change, while the latter state can be reversed by wetting and peeling off the oil in situ by adding nanoemulsion.

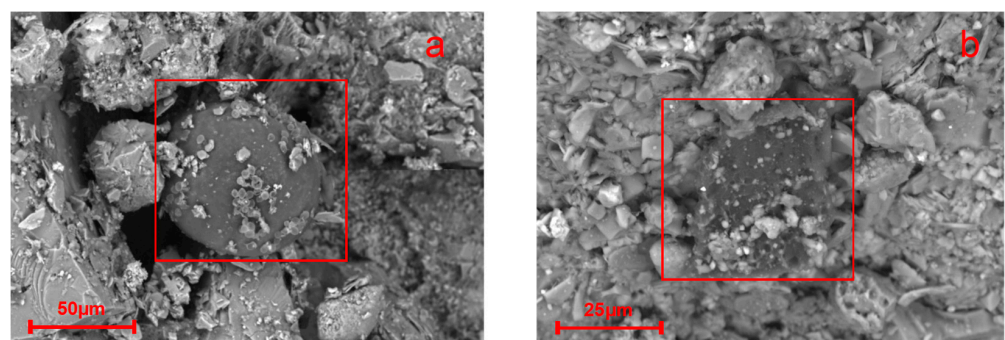


Figure 12. The SEM image of wax precipitation: (a) remaining oil morphology after waterflooding at 25 °C; (b) remaining oil morphology after waterflooding at 65 °C. (The red box shows the shape of the detected crude oil).

The contact angle is measured via the sessile drop method using the JYb-1 contact angle meter at room temperature. The oil contact angle reflects the wetting state of the core samples, with an oil contact angle less than 90° indicating an oil-wet sample and vice versa for a water-wet sample. In Figure 13, a comparison of contact angles before and after aqueous phase injection (hot water of 65 °C and nanoemulsion) is presented. Initially, the measured contact angles of the core were 61.1° and 50.1°, indicating an oil-wet state. However, after hot water flooding and nanoemulsion flooding, the contact angles significantly increased to 122.65° and 127.6°, respectively, indicating an alteration in wettability towards water-wet conditions. The addition of nanoemulsion facilitated the removal of polar components, such as colloid and asphaltene, from the crude oil, weakening the adsorption between the crude oil and rock surfaces [50]. Consequently, the contact angle of the sandstone gradually increased, resulting in a stronger water-wet state. This change in wettability improves the driven pressure of drainage by transforming the capillary pressure from resistance to a driving force.

Furthermore, the oil–water interfacial tensions (IFT) of nanoemulsion and water at different temperatures are measured using the du Noüy ring method. Figure 14 shows the results of the interfacial tension, indicating that the higher the temperature, the lower the interfacial tension, and the addition of nanoemulsion can further reduce the interfacial tension. When the temperature is increased from 45 to 65 °C, the IFT is decreased from 26.38 mN/m to 40.54 mN/m. Compared to this, adding nanoemulsion can effectively reduce the IFT. And the enhancement of IFT reduction increases with the decrease in temperature. This is the reason why nanoemulsions have a better performance when they are used at lower temperatures. The reduction in IFT can decrease the flow resistance in the porous medium and thus mobilize more remaining oil therein. Combining IFT reduction and wettability alteration, nanoemulsion can improve the oil recovery of high-wax oil reservoirs as the temperature gradually decreases to the wax precipitation point.

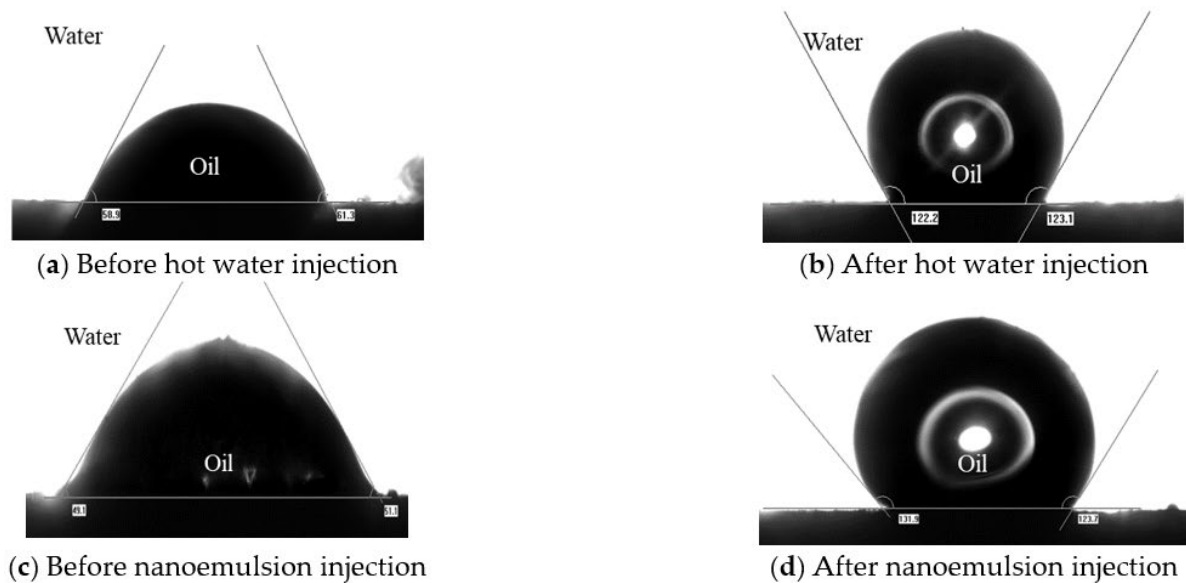


Figure 13. The contact angle images under different injection conditions.

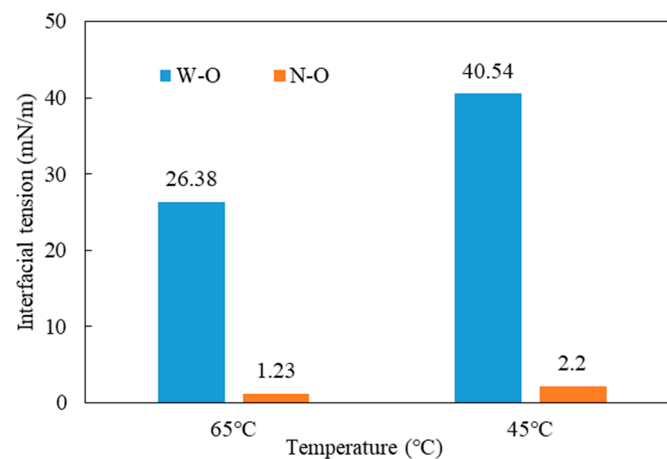


Figure 14. The comparison of interfacial tension under different conditions (W-O represents water–oil; N-O represents nanoemulsion–oil).

5. Conclusions

As wax precipitation from highly waxy crude oil greatly damages reservoir permeability, it has a serious impact on the recovery rate of this type of oil field. Here, we investigated the damage characteristics of wax crystal precipitation from highly waxy crude oil using experimental methods such as rheology, phase penetration, and nuclear magnetism. We also applied SME and wettability experiments to propose anti-waxing and wax suppression methods for improving oil recovery and mitigating the damage in this type of reservoir. The results of the study provide preliminary insights for evaluating the cooling damage of highly waxy oils and provide theoretical support for improving subrecovery. The main conclusions are as follows:

- (1) Zhetybai crude oil mainly consists of light components, based on the analysis of crude oil components. The wax in crude oil is sensitive to temperature, which greatly affects the rheological properties and viscosity of crude oil with temperature changes. The viscosity of crude oil varies greatly with temperature. The rheology of crude oil is also affected by temperature. As the temperature drops below the wax evolution point, the crude oil changes from a Newtonian fluid to a non-Newtonian fluid.

- (2) The cooling damage test results indicate that the reduction in temperature causing wax precipitation in highly waxy crude oil, in turn, impacts the single-phase and two-phase drainage capacity of low-permeability porous media. Specifically, when the temperature is lower than the wax precipitation point, the single-phase permeability of the core decreases as the injection volume increases and the injection temperature decreases. Additionally, the relative permeability curve of oil and water demonstrates that as the temperature decreases, the co-drainage area for the two phases of oil and water also decreases, resulting in an increase in residual oil saturation and a decrease in final oil recovery.
- (3) Based on NMR scanning, the degree of production of highly waxy crude oil in different pores was investigated in relation to temperature and displacement fluid type. With increasing temperature, greater utilization of oil in large pores was observed. Increasing the concentration of nanoemulsion significantly enhanced crude oil production in large pores. In contrast, extraction of crude oil in small pores was challenging due to pore throat obstruction caused by wax deposition. Wax deposition primarily occurs in the pore walls of large pores, allowing for some seepage flow. In contrast, small pores are prone to complete blockage, leading to loss of flow capability. Furthermore, the resulting complex wax deposition can be observed through MRI images. The pattern of remaining oil distribution after displacement can be divided into three types: wax deposition of remaining oil, weak water washing of remaining oil, and unused remaining oil.
- (4) Improving the recovery efficiency of highly waxy crude oil relies on wettability alteration and interfacial tension reduction mechanisms. Increasing the temperature of the water injection can alter the core wettability. Additionally, the nanoemulsion exhibits a stronger capability of reducing interfacial tension, thereby enhancing its effectiveness in enhanced oil recovery (EOR) at low temperatures.

However, it is important to acknowledge that our work has limitations, as the cores used in this study are idealized simulations of the permeability environment of actual reservoirs. The heterogeneity of the reservoir, depositional rhythm, and pore distribution in real cores can all affect wax deposition and flow channel formation to different extents. Wax deposition and its effects in porous media are highly complex processes, influenced by various factors such as temperature, pressure, and crude oil composition. This work can provide valuable insights for the development of high-wax oil reservoirs, but these questions can be further addressed in future research.

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Conflicts of Interest: Authors Xuanran Li, Lun Zhao, Jincai Wang, Shanglin Liu, and Shujun Han are employed by the Research Institute of Petroleum Exploration & Development; Author Minghui Li is employed by the company Oil & Gas and New Energy Branch, PetroChina Corporation; The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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